

Snakes and Ladders

Abdullah Rizwan

2 December, 2024

Abstract

In this report, the average number of rolls for a game of snakes and ladders for a single player was analysed using simulations in python, and compared to the analytical solution given for the game using Absorbing Markov Chains. The average number of rolls using both approaches was found to be $\bar{x} \simeq 36$.

1 Introduction

Snakes and ladders is a classic board game that serves as an excellent model to explore stochastic processes in games. We shall examine the game using python simulations and Absorbing Markov Chain theory to get an analytical result, to analyse the statistical distribution of game lengths. We shall also analyse the impact of the snakes and ladders on the game.

The snakes and ladders board is a 10×10 board with squares 1 to 100. Heads of Snakes drop you down from a higher numbered square to a lower square, and bases of ladders do the opposite. The player progresses by rolling a six sided die, moving forward by the number of spaces indicated. The rules of the game are given as follows:

- The player begins at position 0.
- The player must reach or pass square 100.
- **Ladders:**
 - * $1 \rightarrow 38, 4 \rightarrow 14, 9 \rightarrow 31, 21 \rightarrow 42, 28 \rightarrow 84$
 - * $36 \rightarrow 44, 51 \rightarrow 67, 71 \rightarrow 91, 80 \rightarrow 100$
- **Snakes:**
 - * $16 \rightarrow 6, 47 \rightarrow 26, 49 \rightarrow 11, 56 \rightarrow 53$
 - * $62 \rightarrow 19, 64 \rightarrow 60, 87 \rightarrow 24, 93 \rightarrow 73$
 - * $95 \rightarrow 75, 98 \rightarrow 78$

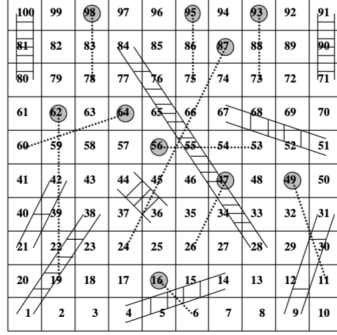


Figure 1: Snakes and ladders board

2 Background theory

We have used Absorbing Markov Chains for this project. Each square on the board corresponds to a state. A transition matrix represents the probability of transition from one state to another.

- Each element P_{ij} of the matrix represents the probability of transition from state i to state j .
- For a snakes and ladders board, our transition matrix is a 101×101 matrix representing the transition probabilities of squares 0 to 100.
- The transition matrix accounts for:
 - Dice rolls from 1 to 6.
 - Jumps due to snakes and ladders.
- The transition matrix will look like:

$$\mathbf{P}_{101 \times 101} = \begin{array}{c|cccccccccccccccccccc} & 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 & \dots & 14 & \dots & 38 & \dots & 78 & \dots & 98 & 99 & 100 \\ \hline 0 & 0 & 0 & \frac{1}{6} & \frac{1}{6} & 0 & \frac{1}{6} & \frac{1}{6} & 0 & \dots & \frac{1}{6} & \dots & \frac{1}{6} & \dots & 0 & \dots & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \dots & 0 & \dots & 0 & \dots & 0 & \dots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 97 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \dots & 0 & \dots & 0 & \dots & \frac{1}{6} & \dots & 0 & \frac{1}{6} & \frac{4}{6} \\ 98 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \dots & 0 & \dots & 0 & \dots & 0 & \dots & 0 & 0 & 0 \\ 99 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \dots & 0 & \dots & 0 & \dots & 0 & \dots & 0 & 0 & 1 \\ 100 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \dots & 0 & \dots & 0 & \dots & 0 & \dots & 0 & 0 & 1 \end{array}$$

The squares are split into 2 classes: Transient states, and absorbing states. If the game lands on an absorbing state, it ends, otherwise it proceeds according to the transition matrix. If there are T transition states and A absorbing states, then the transition matrix P is split as [1]:

$$P = \begin{bmatrix} Q & 0 \\ R & I_A \end{bmatrix}$$

Where $Q \in \mathbb{R}^{T \times T}$, I_A is an $A \times A$ identity matrix, $R \in \mathbb{R}^{A \times T}$, and 0 is a $T \times A$ matrix of zeroes.

Now:

- Transient states: Squares 0 to 99.
- Absorbing state: Square 100.

The expected number of moves to go from square 0 to the absorbing state, square 100 is then given by:

$$\langle n \rangle = \sum_{n=1}^{\infty} nP(n) = (I - Q)^{-1} \quad (1)$$

$$\langle n^2 \rangle = \sum_{n=1}^{\infty} n^2 P(n) = (I - Q)^{-2} (I + Q) \quad (2)$$

Now, relabelling $(I - Q)^{-1}$ as N , which also gives us $(I - Q)^{-2} (I + Q) = 2N^2 - N$.

The expected number of visits to each transient site before absorption is given by:

$$\sum_{i \in T} e_i N = \mathbf{1}^T N \equiv t^T$$

The i^{th} element of this vector represents the expect number of visits to the i^{th} site before absorption. The variance is given by:

$$var(t^T) = \underbrace{2\mathbf{1}^T N^2 - t^T}_{\langle n \rangle} - \underbrace{(t^T \odot t^T)}_{\langle n^2 \rangle}$$

3 Methods and algorithms

3.1 Algorithm

The simulation model works by simulating dice rolls by iterating over 10000 games. The simulation works as follows:

- The simulation function inputs the dictionaries containing snake and ladder positions, and number of simulations.
- Start a for loop for given number of simulations.
- Set position to 0.
- Start a while loop while position is less than 100.
- Simulate a random dice roll using a NumPy function.
- Increment position by dice roll.
- If the player lands on a square with a snake head or ladder base, adjust position accordingly.
- Stop if position reaches or exceeds 100.

4 Results

4.1 Simulation Results

The probability distribution and histogram for the number of rolls for 10000 games is given below:

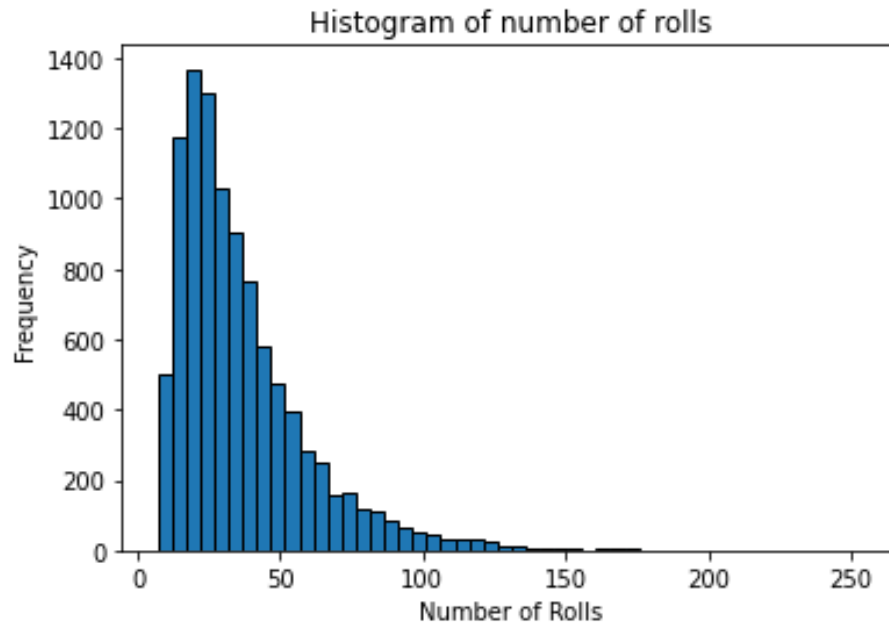


Figure 2: Histogram of rolls for the default board

Average number of rolls according to simulation results : 35.9067

Variance according to simulation results : 559.0845

Standard deviation : 23.6449

Median : 29.0

Mode: 23.0

Shortest game : 7

Longest game : 229

10th Percentile : 14

25th Percentile : 20

50th Percentile : 29

75th Percentile : 45

90th Percentile : 66

The cumulative probability density function is given as follows:

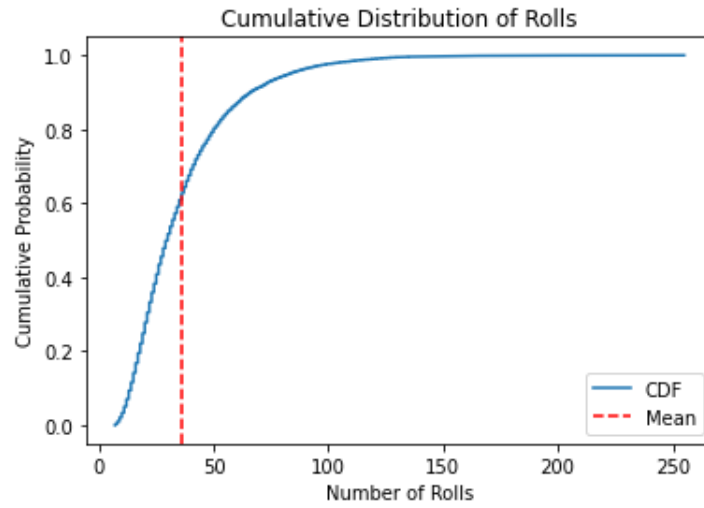


Figure 3: Cumulative probability density function for the default board

4.2 Analytical Results

The analytical results from Absorbing Markov Chain theory implemented in python are given below:

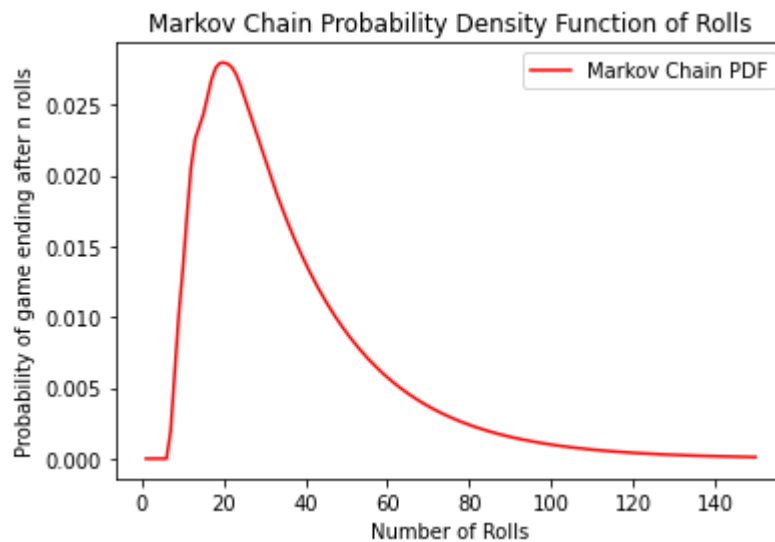


Figure 4: Probability density function for Rolls using Absorbing Markov Chains

Expected number of steps to complete the game(Average rolls): 35.8349
Variance in number of steps to complete the game: 545.4000
Standard deviation: 23.3538

4.3 Analysis

The analytical results using Markov Chains, and the results from the simulation are close. An overlay of the probability density function obtained using Markov Chains on the histogram from the simulation results highlights this:

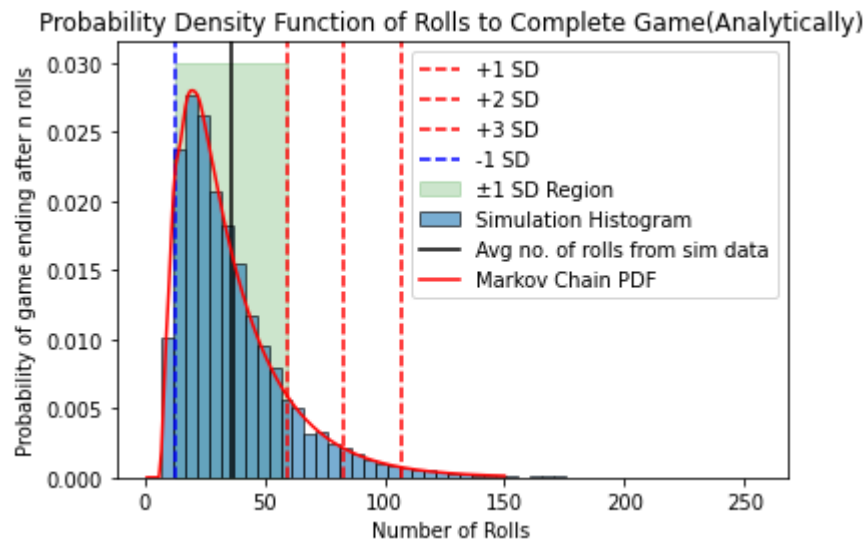


Figure 5: Overlay of Markov Chain PDF onto simulation results

The histogram is a right skewed distribution, with most games completing between 20-40 rolls. The peak is at approximately 23 rolls. It agrees quite closely with the analytical result.

- 50% of the games finish in less than 29 rolls, the majority of games are relatively quick. The average is skewed due to games with more snake encounters lasting much longer.
- 10% of the games finished in less than 14 rolls.
- 10% of the games lasted longer than 66 rolls.
- There were approximately 500 outliers.

4.4 Validation

We find that the errors from the simulation results are:

Absolute errors:

Average number of rolls: 0.0718

Variance: 13.6845

Standard deviation: 0.2912

Relative errors:

Average number of rolls: 0.20026%

Variance: 2.50907%

Standard deviation: 1.24677%

Metric	Simulation Result	Analytical Result	Absolute Error	Relative Error
Average Rolls	35.9067	35.8349	0.0718	0.20026%
Variance	559.0845	545.4000	13.6845	2.50907%
Standard Deviation	23.6449	23.3538	0.2912	1.24677%

Table 1: Comparison of Simulation and Analytical Results

The relative error for the average number of rolls is 0.2%, this implies strong agreement of simulated and analytical results, but the difference in variance is 2.5%. The simulation provides a good estimate for the number of rolls, but the variance and standard deviation are slightly under or overestimated.

Possible causes of error might be:

- Insufficient number of simulations: 10,000 simulations may not be enough to capture the randomness of the game. To rectify this, here are the results with 100,000 simulations:
 - Average number of rolls: 35.89378
 - Variance : 549.21219
 - Standard deviation: 23.43527
 - Errors:
 - Average number of rolls: 0.16420%
 - Variance: 0.69895%
 - Standard deviation: 0.34887%
 - The error in the variance reduced significantly, but the error in the average number of rolls was not affected much.
- Discrete versus continuous model: The simulation is a discrete model, but the analytical solution is a continuous probability function. This might explain the discrepancy in the results for the average number of rolls.

4.5 Analysis of snake and ladder impact

Snakes and ladders significantly impact game duration. Counting the number of snakes and ladders hit for 10,000 simulations and calculating their impact on the players position, we observe that:

- Snakes Encountered: 39593
- Ladders Encountered: 32262
- Average Snake Penalty: 25.97
- Average Ladder Gain: 24.31

We shall also simulate the game with only snakes, and only ladders to further analyse their impact:

4.5.1 Simulation with only ladders

Running 10,000 simulations with only ladders, we have that:

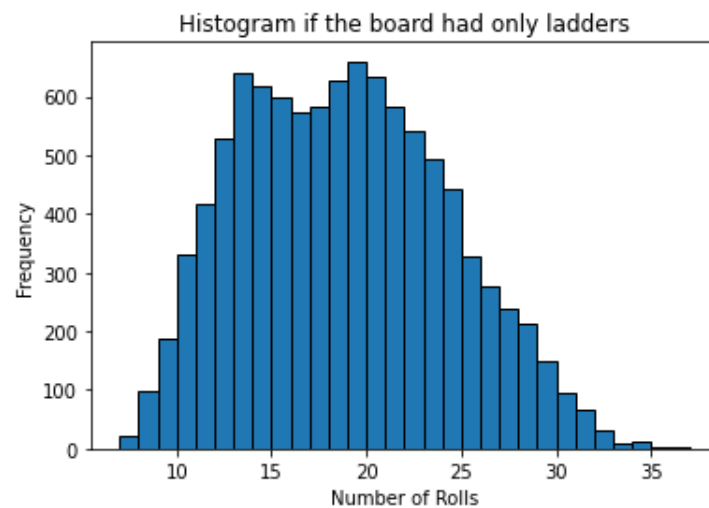


Figure 6: Histogram of number of rolls for a board with only ladders

The distribution is symmetric(roughly).

- Simulation results:
Average number of rolls : 18.3607
Variance : 29.23339
Standard deviation : 5.40679

- Analytical results from Markov Chains:
Average number of rolls : 18.30410
Variance : 29.32683
Standard deviation : 5.41542

The average number of rolls has a significant reduction going from 35.83 to 18.30. The variance also significantly lowers from 545.40 to 29.32 with the standard deviation going from 23.35 to 5.41, implying that ladders both shorten the game and make it more predictable.

Metric	Simulated Value	Analytical Value	Absolute Error	Relative Error (%)
Average Number of Rolls	18.3607	18.3041	0.0566	0.31%
Variance	29.23339	29.32683	0.09344	0.32%
Standard Deviation	5.40679	5.41542	0.00863	0.16%

Table 2: Absolute and Relative Errors between Simulated and Analytical Results for only ladders.

The error in variance also reduces significantly compared to the standard board. Here is an overlay of the analytical and simulated results:

Probability Density Function of Rolls for only ladders to Complete Game(Analytically)

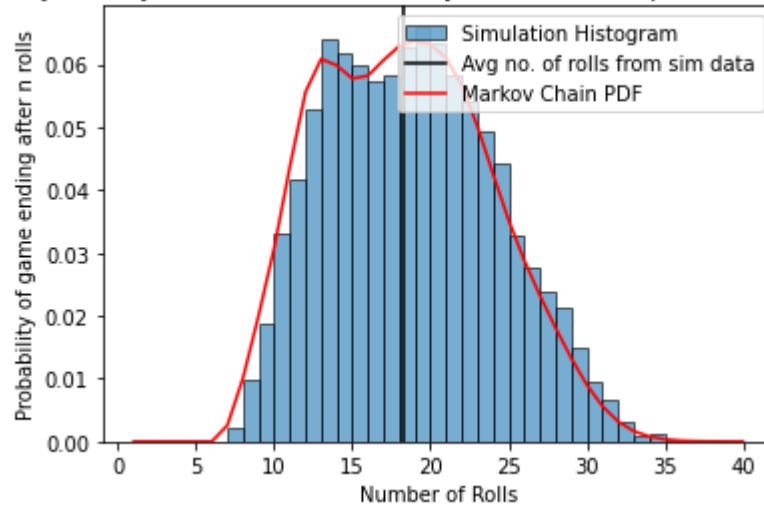


Figure 7: Overlay of Markov PDF and simulated results for a board with only ladders

4.5.2 Simulation with only snakes

For simulation with only snakes we have:

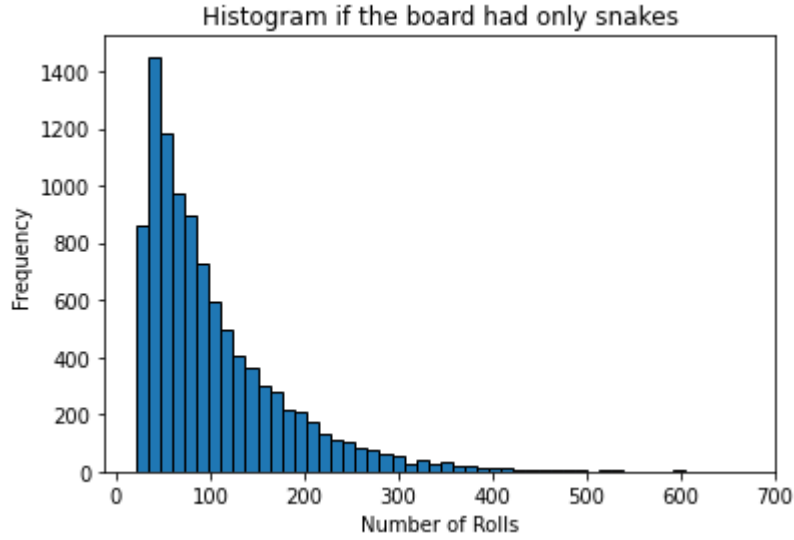


Figure 8: Histogram of rolls with only snakes

- Simulation results:
Average number of rolls : 103.8537
Variance : 5925.02369
Standard deviation : 76.97417
- Analytical results from Markov Chains:
Average number of rolls : 104.64763
Variance: 6090.74568
Standard deviation : 78.04322

The average number of rolls increased from 35.83 to 104.65. The variance increased from 545.40 to 6090.70, with the standard deviation increasing from 23.35 to 78.04.

These facts imply that snakes increase the duration of the game drastically and introduce unpredictability, leading to a broader range of game durations. They introduce setbacks in the position highly sensitive to where they are encountered.

Metric	Simulated Value	Analytical Value	Absolute Error	Relative Error (%)
Average Number of Rolls	103.8537	104.64763	0.79393	0.76%
Variance	5925.02369	6090.74568	165.72199	2.72%
Standard Deviation	76.97417	78.04322	1.06905	1.37%

Table 3: Absolute and Relative Errors between Simulated and Analytical Results for "Snakes Only"

Snakes introduce a higher error in the simulation results, further supporting our conclusion that they introduce more randomness to the game. Here is an overlay of the

analytical and simulated results:

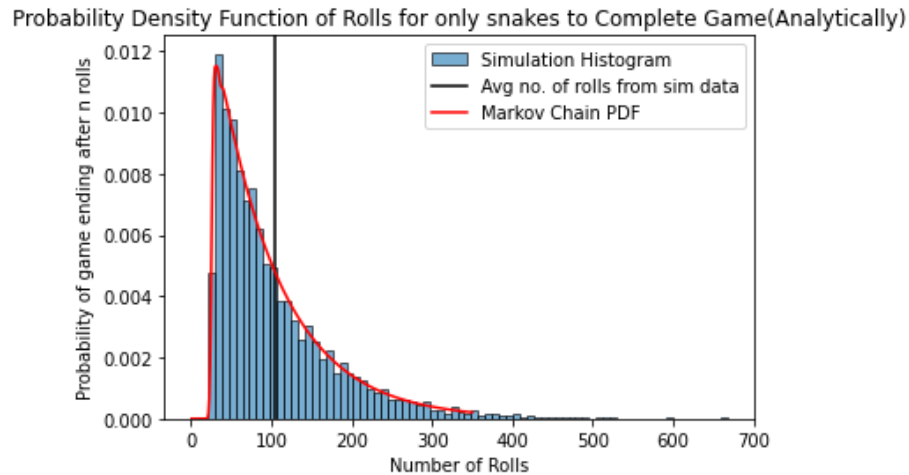


Figure 9: Overlay of Markov PDF and simulation histogram with snakes only

- Simulation results:
Average number of rolls : 103.8537
Variance : 5925.02369
Standard deviation : 76.97417
- Analytical results from Markov Chains:
Average number of rolls : 104.64763
Variance: 6090.74568
Standard deviation : 78.04322

The distribution is right skewed again, to a higher extent than the default board, since longer games are more likely to occur. It has a 'Long tail'.

4.5.3 Simulation without snakes or ladders

Running simulations on a board without snakes or ladders gives:

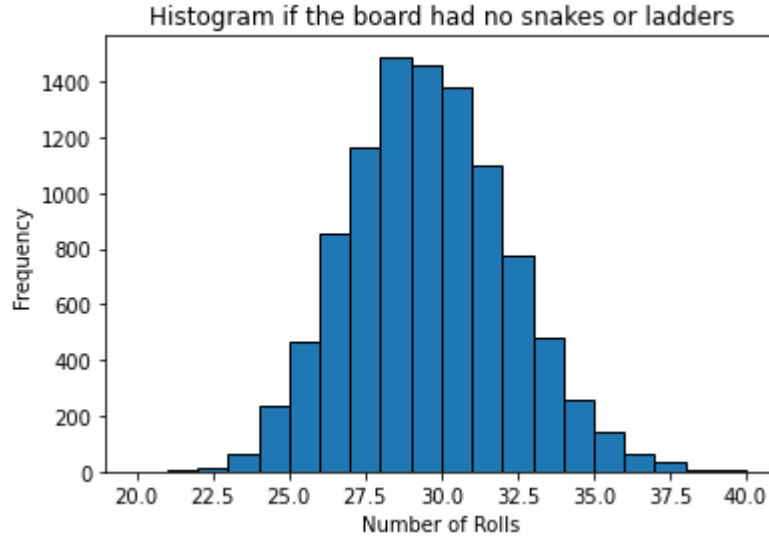


Figure 10: Histogram for empty board(No snakes/ladders)

- Simulation results:
Average number of rolls : 29.0825
Variance : 6.90649
Standard deviation : 2.62802
- Analytical results from Markov Chains:
Average number of rolls : 29.04761
Variance : 6.91609
Standard deviation : 2.62984

The average number of rolls decreased from 35.83 to 29.04. The variance decreased from 545.40 to 6.92, with the standard deviation decreasing from 23.35 to 42.63. This is expected as the removal of snakes and ladders leaves only the randomness of the dice. Indeed, if we were to use 3.5 as the expectation value for the dice, we would have: $\frac{100}{3.5} = 28.57$ rolls, but we can't have a non integer number so its 29.

Metric	Simulated Value	Analytical Value	Absolute Error	Relative Error (%)
Average Number of Rolls	29.0825	29.04761	0.03489	0.1201%
Variance	6.90649	6.91609	0.00960	0.1388%
Standard Deviation	2.62802	2.62984	0.00182	0.0692%

Table 4: Absolute and Relative Errors for a Board Without Snakes or Ladders

The low relative errors confirm that the simulation and analytical results agree. An overlay of the two is given below:

Probability Density Function of Rolls for no snakes/ladders to Complete Game(Analytically)

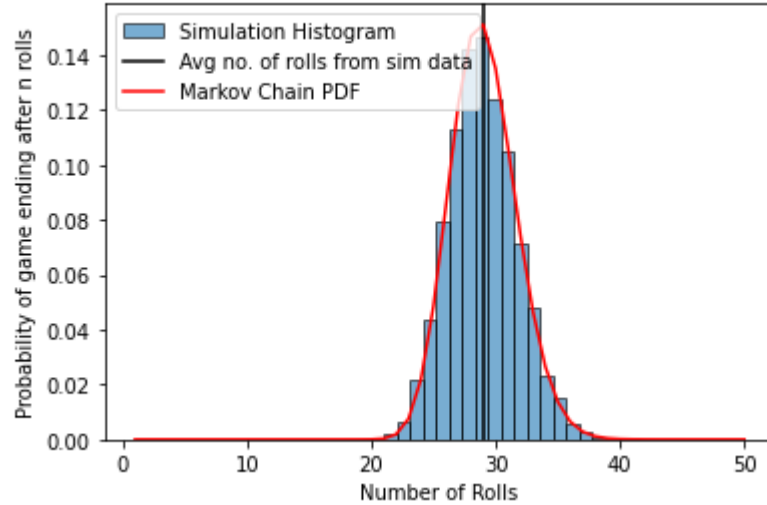


Figure 11: Overlay of Markov PDF and simulated results for a board without snakes or ladders

This distribution is symmetric.

4.6 Heatmap

The heatmap given here indicates the most and least visited squares, showing areas of the board with high traffic. The given figures are normalised.

The top 5 Most Visited squares are:

- Square 44: 2.50%
- Square 26: 2.44%
- Square 84: 2.16%
- Square 11: 1.97%
- Square 100: 1.91%

Top 5 Least Visited Squares (excluding Square 0) are:

- Square 99: 0.38%
- Square 1: 0.39%
- Square 2: 0.39%
- Square 68: 0.45%

- Square 3: 0.46%

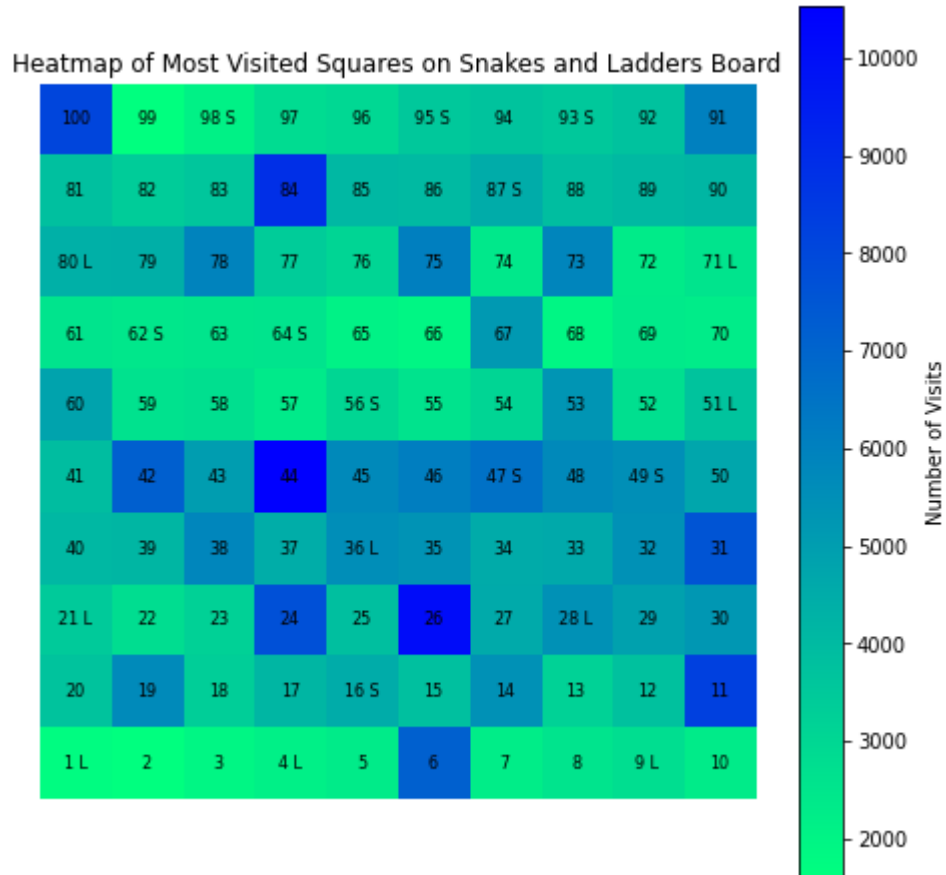


Figure 12: Heatmap of most visited squares during simulation

As expected, squares at snake tails, or ladder heads are visited more than normal squares.

The most visited square is 44, due to its proximity with snake tails and ladder heads, for instance the ladder at 36 leads to 44. Squares like 26, 84 and, 11 are also snake tails or ladder heads, often in proximity with others, leading to higher visits.

The squares at the beginning like squares 1 to 5 are not visited much, as no snake takes the player back here, so the player visits these only once per game and may skip over them entirely, by contrast square 6 is relatively more visited because of the snake head at 16, taking players back to 6.

The squares 65, 66, and, 69 are also not visited much because of their position between two areas highly influenced by snakes and ladders. 67 is visited due to being a ladder head.

Square 99 is not visited much due to it being close to the end, leading to it getting skipped as most players proceed directly to 100 from previous squares like 94, 96, and 97.

5 Further Work

The computational efficiency could be increased by doing the snake and ladder analysis at the same time as running the initial simulations. Further work might be done by:

- Randomizing the positions of the snakes and ladders on the board.
- Changing the ratio of snakes and ladders to see how the game length is affected.
- Using non standard dice eg. Weighted dice, 4 sided dice, etc.
- Changing the board size to see the impact of small changes on the game length.
- Introducing multiple players.
- Changing win conditions, for example, the player must land exactly on 100 to win.

References

- [1] Snakes and ladders resource, given in the project resources.